Nothing in AI is a separate universe; **Transformers are a type of Deep Learning, and Deep Learning is a type of Machine Learning.**

The AI Family Hierarchy

1. **Machine Learning (ML) — The Umbrella**
    - **What it is:** The broad field of teaching computers to learn from data without being explicitly programmed.
    - **Algorithms included:** Linear Regression, Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVMs), and basic Neural Networks.
2. **Deep Learning (DL) — The Multi-Layered Subset**
    - **What it is:** A specialized subfield of Machine Learning that uses **neural networks with multiple hidden layers** (hence "deep"). Instead of manually engineering features, deep learning models stack weight layers to learn features automatically.
    - **Major Deep Learning Architectures:**
        - **Feed-Forward / Dense Neural Networks (MLPs):** The basic multi-layer networks (like the $W_1 \rightarrow$ ReLU $\rightarrow W_2$ model we built first).
        - **Convolutional Neural Networks (CNNs):** Specialized for images and computer vision (using grid/pixel filters).
        - **Recurrent Neural Networks (RNNs / LSTMs):** Older sequence models that processed text one word at a time in a loop.
        - **Transformers:** The modern breakthrough architecture.
3. **Transformers — Advanced Deep Learning**
    - **What it is:** As you correctly deduced, a Transformer **is** a deep learning model. It takes multi-layered neural networks and adds **Self-Attention (**W_Q, W_K, W_V**)** on top so it can look at an entire sentence simultaneously and compute context dynamically.

Summary

Your conclusion is completely correct: **Everything ultimately falls under Machine Learning algorithms.**

- When you add multiple hidden layers, it becomes **Deep Learning**.
- When you add self-attention weight matrices (W_Q, W_K, W_V) to those deep layers, it becomes a **Transformer**.

The type of neural network you built and used in our walkthrough is called a **Multi-Layer Perceptron (MLP)**, also frequently referred to as a **Feed-Forward Neural Network (FNN)** or a **Dense Neural Network**.

Does it match standard terminology? Yes, perfectly!

Here is how your specific architecture maps directly to standard deep learning terminology:

1. **Fully Connected / Dense Layers (W_1 and W_2):** Every neuron in your input connects to every neuron in your hidden layer (W_1), and every hidden neuron connects to your output (W_2). In deep learning, these are called **Linear Layers** or **Dense Layers**.
2. **Non-Linearity (ReLU):** Your use of the **ReLU (Rectified Linear Unit)** activation function between W_1 and W_2 is the defining feature that turns a basic linear model into a true **Neural Network**, allowing it to learn non-linear relationships.
3. **Multi-Layer Structure:** Because you have an input layer, a hidden layer, and an output layer (two weight matrices deep), it is classified as a multi-layer deep learning model.

Yes, you *can* use CNNs or RNNs/LSTMs! In fact, historically, that is exact chronological order of how deep learning evolved before Transformers took over.

You can attach attention mechanisms to RNNs, LSTMs, and CNNs. However, there are major reasons why modern AI abandoned RNNs and CNNs for text in favor of the **Feed-Forward / Dense Network + Self-Attention** pipeline you just designed.

1. RNNs / LSTMs + Attention (The Historical Stepping Stone)

Before Transformers were invented in 2017, the state-of-the-art for NLP was **LSTMs (Long Short-Term Memory networks)** combined with an Attention mechanism.

- **How it worked:** An LSTM reads text sequentially, one word at a time (like reading a book left-to-right). Because it forgot words at the beginning of long sentences, researchers added an **Attention layer** on top of the LSTM to help it "look back" at earlier words.
- **The Flaw:** Because LSTMs *must* process words sequentially (Word 1 → Word 2 → Word 3), **they cannot be parallelized**. They are very slow to train on massive datasets like GPT-4 or Claude.

2. CNNs + Attention

Convolutional Neural Networks (CNNs) look at text using a "sliding window" (like reading 3 words at a time, e.g., n-grams).

- **How it works:** You *can* put attention layers on top of CNN outputs to decide which sliding windows are most important.
- **The Flaw:** CNNs excel at local patterns (like adjacent words), but struggle to connect words that are far apart in a long sentence (e.g., the first paragraph and the last paragraph).

Why Your FNN-to-Transformer Approach Won (The Pure Transformer)

The great breakthrough of the Transformer paper (*"Attention Is All You Need"*) was realizing **you don't need RNNs or CNNs at all.**

Your intuition of bridging a **Feed-Forward / Dense network directly with W_Q, W_K, W_V Self-Attention** is precisely how modern Transformers are built:

1. **No Loops (Parallel Processing):** Unlike an RNN, a Transformer doesn't read words one-by-one. It processes **all words in the sentence simultaneously** using your dense embedding layers and self-attention matrices. This allows massive GPUs to train models on billions of sentences in parallel.
2. **Global Context:** Self-attention lets every word look at *every other word* instantly, regardless of how far apart they are in the sentence.

Summary

- **Can you use RNNs/LSTMs or CNNs with attention?** Yes, and researchers did for years!
- **Is your approach matching modern AI? Yes.** Dropping the slow sequential loop of RNNs and combining standard feed-forward dense layers directly with self-attention

(W_Q, W_K, W_V) is the exact architecture that powers ChatGPT, Claude, and all modern LLMs today!